

Supplemental material for “Foraging behavior and abundance determine the relative importance of pollinators in apple orchards” by Russo, Blitzer, Park, and Danforth

We identified three traits we felt could directly or indirectly impact per-visit pollen deposition: pollen purity, bee body size, and flower handling behavior. Here in the supplemental materials, we provide a more thorough explanation of pollen purity and bee body size, as well as the relationship between purity, size, and behavior with pollen deposition.

Pollen purity

We define pollen purity as the proportion of the pollen load consisting of apple pollen relative to pollen of other species. Previous studies (e.g. (Sipes and Tepedino, 2005) have defined a pure pollen load as >90% of a single pollen type. Our hypothesis is that pollen purity will be positively correlated with pollen deposition (Larsson, 2005) and we test this by analyzing the pollen carried by 170 bees of 14 bee species and comparing it to a published dataset on pollen deposition in the same system (Park et al., 2015). The 14 species we analyzed represented 99% of total bee functional group diversity (see Table S1 for functional group classification).

We used the methods described in (Edens-Meier et al., 2011) to wash pollen off of net-collected bee specimens (see also (Bernhardt, 1987). The pollen was washed off of bee specimens onto microscope slides, where we counted at least 300 pollen grains from ten randomly selected individuals of each of 14 bee species. Together, these species comprised 85% of the total bee abundance found in the apple orchards during our study. The pollen counts

involved at least 3 random transects across the slide for each individual. Pollen grains were photographed, measured, and categorized using a reference library of pollen collected from all flowering angiosperms at the study sites in 2014 (<http://blogs.cornell.edu/pollengrains/>), as well as several reference books (Crompton and Wojtas, 1993; Kapp et al., 2000; Lewis et al., 1983). *Malus* pollen was identified based on size, shape, surface sculpturing, and the presence/absence of similar species based on extensive floral surveys (Russo *unpublished data*).

We quantified percent *Malus* pollen (pollen purity from here on) for each species as the number of *Malus* pollen grains divided by the total number of pollen grains counted for each individual. Pollen purity is evaluated for its relationship with pollen deposition and seed set in the orchards (see below).

Male bees comprised approximately 12% of the overall bee abundance in the study. For most of the species, females outnumber males, but males outnumbered females significantly in *Andrena crataegi*, *A. imitatrix*, and *Xylocopa virginica* (Table S1). For this reason, we also counted pollen grains on male *A. crataegi* and *X. virginica* (these two species alone comprise 62% of male abundance). If we leave out all bee species for which pollen purity was not assessed directly (i.e. species represented by functional groups) we counted pollen on roughly 83% of total bee abundance. To quantify the services provided by the wild bee community as a whole, we represented species for which we did not measure pollen purity directly by using measured values from species in the same subgenus (placing them in taxonomically and morphologically based functional groups). In comparing our model results with and without the species we did not directly measure, we found the strongest models include functional group representation (Table 2).

Bees carried between 28% (*Lasioglossum hitchensi*) and 80% (*Andrena miserabilis*) *Malus* pollen, on average (Fig. S1, Table 1). While carrying less pollen, males of both *Xylocopa virginica* and *Andrena crataegi* had higher average pollen purity than their female counterparts. When we calculated pollinator importance as the pollen purity multiplied by the abundance of each bee species, wild bees were found to be more important than honeybees across all the orchards, on average, but this difference was not significant ($P > 0.05$, Fig. 2C).

Body size

Attributes of bee morphology might contribute to the amount of pollen deposited per visit. Mean body size varies widely among bee species and could thus affect relative pollen deposition. We hypothesize that larger bees, because of their larger surface area, will deposit more pollen per visit than smaller bees. We test this by comparing body size data (bee length and intertegular distance) from field collected specimens to a pollen deposition study in the same system (Park et al., 2015).

We used the information on bee lengths available in Mitchell (Mitchell, 1962) for all the male and female bee species present in our study. When a range was provided, we used average bee length as a measure of size. In addition, we measured intertegular distances for the male and female bees of the species represented in our study (Table S1).

The length of bees in this study varied between 4 and 22 mm, with an average size of 9.93 mm. The ten most abundant bee species had an average size of 12.07 mm (Supplemental Table 1). When bee length was multiplied by abundance across all species, wild bees were found to be more important than honeybees on average, but this difference was not significant ($P >$

0.05, Fig. 2D). The intertegular distances of bees varied between 0.81 and 6.41 mm, with an average of 2.18. The intertegular distance correlated strongly with bee length (0.88, $P < 0.01$, Fig. S2), and behaved similarly in all analyses; thus, the results based on intertegular distances are presented in the supplemental materials.

Relationship between pollen purity, size, and behavior with per-visit pollen deposition

To test which factors are good proxies for the pollen bees deposit on flower stigmas, we compared our data to per-visit pollen deposition data from a recent study by Park *et al.* (2015). Park *et al.* (2015) conducted a study of pollen deposition in apple orchards involving three different bee species. First, they interviewed **individual bees** to quantify the number of pollen grains deposited on apple stigmas after a single visit by bees of the subgenus *Melandrena*, bees of the genus *Bombus*, or *Apis mellifera* (see Park *et al.* 2015 for further details). They also captured bees, chilled them, and applied either the scopa or the thorax of the bees to stigmas and then allowed the fruit to develop. This second part of the study was conducted because flowers are removed from stems in the interview method and it is not possible to compare pollen deposition to fruit set using that methodology. Park *et al.* (2015) used chilled bees of *Andrena* subgenus *Melandrena* (comprising several, closely related species), *Apis mellifera*, and *Bombus impatiens*. They found a strong, positive correlation between pollen deposition and fruit set (see Park *et al.* 2015). The authors also have unpublished data on the pollen deposition of interviewed *Xylocopa virginica*, and chilled *Andrena nasonii* (Park, *unpub data*), which we also compare here to our measures of pollen purity.

Our results suggest the pollen purity of the bees does relate to the pollen deposited on the stigmas, but that body size mediates this relationship in the field. For example, there was a direct correlation between pollen purity and pollen deposition for chilled bees applied to stigmas (estimate 1.6, $P < 0.001$, Fig. S2). However, for live interviewed bees, pollen deposition was positively correlated with the product of bee size and purity (estimate 103.25, $P < 0.01$, Fig. S2), although neither pollen purity nor bee size alone predicted pollen deposition.

In addition to the individual contributions of these factors, they might interact in important ways. For example, the relative contribution of these two components might be driven by scale. One might expect that per-visit effectiveness is more critical at the scale of an individual flower (Larsson, 2005), while abundance dominates at larger scales (Olsen, 1996; Winfree et al., 2015).

At the scale of an individual flower, the combined results of Park *et al.* (2015) and the pollen purity analysis show that pollen deposition is correlated with pollen purity when chilled bees are applied to the stigma of apple flowers, but that this effect is mediated by bee size in live foraging bees. Thus, pollen purity is not an important factor at the scale of the entire orchard. The distinction between pollinator importance at these two different scales is important for understanding how factors that increase the probability of a single flower setting fruit relate to factors that increase orchard-level yield. Our results demonstrate that these factors are not necessarily the same.

Table S1. A list of bee species found in the orchards during the time of the seed set study, including the purity of their pollen load, functional group assignments, pollen density score (on a scale from 1 to 10), body length, reported proportion of top working from Martins et al (2015) and Park et al (2015), the floral handling values we used, and the Intertegumental distance on our measured specimens, as well as their abundance.

	Species	Functional Group	Pollen Purity	Score	Length (mm)	Martins et al 2015	Park et al 2015	Floral Handling	ITD	Abundance
Female	<i>Agapostemon sericeus</i>	<i>Halictus rubicundus</i>	0.65	6.23	11	na	na	na	2.7	4
	<i>Andrena barbilabris</i>	<i>Andrena regularis/vicina</i>	0.72	7.20	10.5	1.00	na	1.00	2.0345	6
	<i>Andrena carlini</i>	<i>Andrena regularis/vicina</i>	0.72	7.20	12.5	1.00	na	1.00	3.425	66
	<i>Andrena commoda</i>	<i>Andrena regularis/vicina</i>	0.72	7.20	11	1.00	na	1.00	2.2	3
	<i>Andrena crataegi</i>	<i>Andrena crataegi</i>	0.48	4.01	11.5	1.00	na	1.00	2.2	232
	<i>Andrena cressonii</i>	<i>Andrena hippotes</i>	0.51	5.12	9.5	0.97	na	0.97	2.21975	31
	<i>Andrena dunningi</i>	<i>Andrena regularis/vicina</i>	0.72	7.20	11.5	1.00	na	1.00	2.4375	16
	<i>Andrena erythronii</i>	<i>Andrena regularis/vicina</i>	0.72	7.20	12	1.00	na	1.00	2.29775	4
	<i>Andrena forbesii</i>	<i>Andrena hippotes</i>	0.51	5.12	9.5	0.97	na	0.97	2.0995	45

	<i>Andrena hippotes</i>	<i>Andrena hippotes</i>	0.51	5.1 2	9.5	0.97	na	0.97	2	130
	<i>Andrena imitatrix</i>	<i>Andrena hippotes</i>	0.51	5.1 2	10	1.00	na	0.97	1.764	10
	<i>Andrena mandibularis</i>	<i>Andrena regularis/vicina</i>	0.72	7.2 0	9.5	0.97	na	0.97	1.88	18
	<i>Andrena milwaukeeensis</i>	<i>Andrena regularis/vicina</i>	0.72	7.2 0	11	1.00	na	1.00	2.289 42	10
	<i>Andrena miserabilis</i>	<i>Andrena miserabilis</i>	0.80	7.7 5	8.2	0.97	na	0.97	2.208	149
	<i>Andrena morrisonella</i>	<i>Andrena hippotes</i>	0.51	5.1 2	9	0.97	na	0.97	1.764	13
	Species	Functional Group	Pollen Purity	Sco re	Length (mm)	Martins et al 2015	Park et al 2015	Floral Handling	ITD	Abunda nce
	<i>Andrena nasonii</i>	<i>Andrena nasonii</i>	0.68	6.2 2	8	0.97	na	0.97	2.183	96
	<i>Andrena nivalis</i>	<i>Andrena regularis/vicina</i>	0.72	7.2 0	13	1.00	na	1.00	2.621 08	1
	<i>Andrena nuda</i>	<i>Andrena hippotes</i>	0.51	5.1 2	9	0.97	na	0.97	2.003	1
	<i>Andrena perplexa</i>	<i>Andrena regularis/vicina</i>	0.72	7.2 0	13.5	1.00	na	1.00	2.579 22	46
	<i>Andrena pruni</i>	<i>Andrena</i>	0.72	7.2	10.25	1.00	na	1.00	2.295	33

		<i>regularis/vicina</i>		0					86	
	<i>Andrena regularis</i>	<i>Andrena regularis</i>	0.74	7.4 2	12	1.00	0.75	0.81	2.6	291
	<i>Andrena rugosa</i>	<i>Andrena hippotes</i>	0.51	5.1 2	9	0.97	na	0.97	1.915	47
	<i>Andrena vicina</i>	<i>Andrena vicina</i>	0.70	6.9 7	12.5	1.00	0.75	0.81	2.440 76	182
	<i>Andrena w-scripta</i>	<i>Andrena nasonii</i>	0.68	6.2 2	9.5	1.00	na	0.97	1.633	10
	<i>Apis mellifera</i>	<i>Apis mellifera</i>	0.70	4.1 5	12.5	0.62	0.47	0.54	3.3	2610
	<i>Augochlora pura</i>	<i>Augochlora pura</i>	0.45	2.4 1	8	na	na	na	2.233	44
	<i>Augochlorella aurata</i>	<i>Augochlora pura</i>	0.45	2.4 1	5.5	na	na	na	1.512	6
	<i>Augochloropsis metallica</i>	<i>Augochlora pura</i>	0.45	2.4 1	9	na	na	na	2.642	1
	<i>Bombus bimaculatus</i>	<i>Bombus impatiens</i>	0.61	4.0 8	19.5	1.00	1	1.00	3.544	25
	<i>Bombus fervidus</i>	<i>Bombus impatiens</i>	0.61	4.0 8	20	1.00	1	1.00	5.568	1
	<i>Bombus griseocollis</i>	<i>Bombus impatiens</i>	0.61	4.0 8	22	1.00	1	1.00	4.312	15

	<i>Bombus impatiens</i>	<i>Bombus impatiens</i>	0.61	4.08	19	1.00	1	1.00	3.448	181
	Species	Functional Group	Pollen Purity	Score	Length (mm)	Martins et al 2015	Park et al 2015	Floral Handling	ITD	Abundance
	<i>Bombus perplexus</i>	<i>Bombus impatiens</i>	0.61	4.08	19	1.00	1	1.00	5.8	14
	<i>Bombus ternarius</i>	<i>Bombus impatiens</i>	0.61	4.08	18	1.00	1	1.00	4.804	4
	<i>Bombus terricola</i>	<i>Bombus impatiens</i>	0.61	4.08	18	1.00	1	1.00	5.978	1
	<i>Ceratina calcarata</i>	na	na	na	na	0.95	na	0.95	1.4	15
	<i>Ceratina dupla</i>	na	na	na	na	0.95	na	0.95	1.4	1
	<i>Colletes inaequalis</i>	<i>Colletes inaequalis</i>	0.54	5.36	12.5	1.00	na	1.00	2.878	41
	<i>Halictus confusus</i>	<i>Halictus rubicundus</i>	0.65	6.23	7	0.95	na	0.95	1.883	16
	<i>Halictus rubicundus</i>	<i>Halictus rubicundus</i>	0.65	6.23	10.5	0.95	na	0.95	2.65	16
	<i>Lasioglossum abanci</i>	<i>Lasioglossum hitchensi</i>	0.28	1.63	6	0.95	na	0.95	1.092	2
	<i>Lasioglossum cinctipes</i>	<i>Lasioglossum hitchensi</i>	0.28	1.63	8	0.95	na	0.95	2	15

	<i>Lasioglossum coeruleum</i>	<i>Lasioglossum hitchensi</i>	0.28	1.6 3	7	0.95	na	0.95	1.288	5
	<i>Lasioglossum coriaceum</i>	<i>Lasioglossum hitchensi</i>	0.28	1.6 3	9	0.95	na	0.95	2.708	2
	<i>Lasioglossum cressonii</i>	<i>Lasioglossum hitchensi</i>	0.28	1.6 3	6.5	0.95	na	0.95	1.623	8
	<i>Lasioglossum ephialtum</i>	<i>Lasioglossum hitchensi</i>	0.28	1.6 3	5.5	0.95	na	0.95	1.22	1
	<i>Lasioglossum foxii</i>	<i>Lasioglossum hitchensi</i>	0.28	1.6 3	6.5	0.95	na	0.95	1.2	20
	<i>Lasioglossum heterognathum</i>	<i>Lasioglossum hitchensi</i>	0.28	1.6 3	4	0.95	na	0.95	0.96	1
	<i>Lasioglossum hitchensi</i>	<i>Lasioglossum hitchensi</i>	0.28	1.6 3	5.185	0.95	na	0.95	1.1	55
	Species	Functional Group	Pollen Purity	Sco re	Length (mm)	Martins et al 2015	Park et al 2015	Floral Handling	ITD	Abunda nce
	<i>Lasioglossum imitatum</i>	<i>Lasioglossum hitchensi</i>	0.28	1.6 3	4.18	0.95	na	0.95	0.814	1
	<i>Lasioglossum laevissimum</i>	<i>Lasioglossum hitchensi</i>	0.28	1.6 3	6	0.95	na	0.95	1.617	8
	<i>Lasioglossum lineatulum</i>	<i>Lasioglossum hitchensi</i>	0.28	1.6 3	6	0.95	na	0.95	1.692	15

	<i>Lasioglossum nigroviride</i>	<i>Lasioglossum hitchensi</i>	0.28	1.6 3	7	0.95	na	0.95	1.5	1
	<i>Lasioglossum nymphaearum</i>	<i>Lasioglossum hitchensi</i>	0.28	1.6 3	7	0.95	na	0.95	2.092	1
	<i>Lasioglossum obscurum</i>	<i>Lasioglossum hitchensi</i>	0.28	1.6 3	6	0.95	na	0.95	1.1	8
	<i>Lasioglossum paradmiraandum</i>	<i>Lasioglossum hitchensi</i>	0.28	1.6 3	5.5	0.95	na	0.95	1.1	11
	<i>Lasioglossum perpunctatum</i>	<i>Lasioglossum hitchensi</i>	0.28	1.6 3	6	0.95	na	0.95	1.2	1
	<i>Lasioglossum planatum</i>	<i>Lasioglossum hitchensi</i>	0.28	1.6 3	5.685	0.95	na	0.95	1.1	3
	<i>Lasioglossum quebecense</i>	<i>Lasioglossum hitchensi</i>	0.28	1.6 3	7	0.95	na	0.95	1.278	16
	<i>Lasioglossum truncatum</i>	<i>Lasioglossum hitchensi</i>	0.28	1.6 3	7	0.95	na	0.95	1.648	4
	<i>Lasioglossum versans</i>	<i>Lasioglossum hitchensi</i>	0.28	1.6 3	6	0.95	na	0.95	1.2	1
	<i>Lasioglossum versatum</i>	<i>Lasioglossum hitchensi</i>	0.28	1.6 3	6	0.95	na	0.95	1.212	18
	<i>Lasioglossum weemsi</i>	<i>Lasioglossum hitchensi</i>	0.28	1.6 3	4.5	0.95	na	0.95	1.12	6
	<i>Lasioglossum zonulum</i>	<i>Lasioglossum</i>	0.28	1.6	9	0.95	na	0.95	2.742	9

		<i>hitchensi</i>		3						
	<i>Nomada cressonii</i>	na	na	na	na	na	na	na	1.45	3
	<i>Nomada ovata</i>	na	na	na	na	na	na	na	1.56	1
	Species	Functional Group	Pollen Purity	Score	Length (mm)	Martins et al 2015	Park et al 2015	Floral Handling	ITD	Abundance
	<i>Nomada pygmaea</i>	na	na	na	na	na	na	na	1.905	1
	<i>Osmia bucephala</i>	<i>Osmia cornifrons</i>	0.76	7.10	15.5	na	na	na	3.544	2
	<i>Osmia conjuncta</i>	<i>Osmia cornifrons</i>	0.76	7.10	9	na	na	na	2.26	2
	<i>Osmia cornifrons</i>	<i>Osmia cornifrons</i>	0.76	7.10	14	na	na	na	2.9	30
	<i>Osmia pumila</i>	<i>Osmia cornifrons</i>	0.76	7.10	8	na	na	na	1.8515	4
	<i>Xylocopa virginica</i>	<i>Xylocopa virginica</i>	0.61	6.08	21	na	0.82	0.82	5.6832	31
Male	<i>Andrena carlini</i>	<i>Andrena regularis/vicina</i>	0.72	7.20	12	1.00	na	1.00	2.085	5
	<i>Andrena commoda</i>	<i>Andrena regularis/vicina</i>	0.72	7.20	10.5	1.00	na	1.00	1.709	16
	<i>Andrena crataegi</i>	male <i>Andrena crataegi</i>	0.70	1.69	10.5	1.00	na	1.00	1.779	466

	<i>Andrena cressonii</i>	<i>Andrena hippotes</i>	0.51	5.1 2	9	0.97	na	0.97	1.628 25	3
	<i>Andrena forbesii</i>	<i>Andrena hippotes</i>	0.51	5.1 2	8.75	0.97	na	0.97	1.657 5	12
	<i>Andrena geranii</i>	<i>Andrena nasonii</i>	0.68	6.2 2	8	0.97	na	0.97	1.44	3
	<i>Andrena hippotes</i>	<i>Andrena hippotes</i>	0.51	5.1 2	8.5	0.97	na	0.97	1.607	33
	<i>Andrena imitatrix</i>	<i>Andrena hippotes</i>	0.51	5.1 2	8.5	0.97	na	0.97	1.423 2	29
	<i>Andrena mandibularis</i>	<i>Andrena regularis/vicina</i>	0.72	7.2 0	8.5	0.97	na	0.97	1.346	1
	<i>Andrena miserabilis</i>	<i>Andrena miserabilis</i>	0.80	7.7 5	7	0.97	na	0.97	1.792	17
	<i>Andrena morrisonella</i>	<i>Andrena hippotes</i>	0.51	5.1 2	7	0.97	na	0.97	1.423 2	3
	Species	Functional Group	Pollen Purity	Sco re	Length (mm)	Martins et al 2015	Park et al 2015	Floral Handling	ITD	
	<i>Andrena nasonii</i>	<i>Andrena nasonii</i>	0.68	6.2 2	7	0.97	na	0.97	1.758	88
	<i>Andrena perplexa</i>	<i>Andrena regularis/vicina</i>	0.72	7.2 0	11.5	1.00	na	1.00	1.935 22	21
	<i>Andrena pruni</i>	<i>Andrena</i>	0.72	7.2	8.5	0.97	na	0.97	1.628	3

		<i>regularis/vicina</i>		0					05	
	<i>Andrena regularis</i>	<i>Andrena regularis</i>	0.74	7.4 2	10.3	1.00	0.75	0.81	1.734	1
	<i>Andrena rugosa</i>	<i>Andrena hippotes</i>	0.51	5.1 2	8	0.97	na	0.97	2.125	8
	<i>Andrena vicina</i>	<i>Andrena vicina</i>	0.70	6.9 7	11	1.00	0.75	0.81	1.816 08	14
	<i>Andrena wilkella</i>	<i>Andrena hippotes</i>	0.51	5.1 2	10	1.00	na	1.00	3.042	1
	<i>Andrena w-scripta</i>	<i>Andrena nasonii</i>	0.68	6.2 2	8.5	0.97	na	0.97	1.421	30
	<i>Ceratina calcarata</i>	na	na	na	na	0.95	na	0.95	1.269 31	31
	<i>Nomada cressonii</i>	na	na	na	na	na	na	na	1.41	17
	<i>Nomada luteoloides</i>	na	na	na	na	na	na	na	1.57	4
	<i>Nomada media</i>	na	na	na	na	na	na	na	na	1
	<i>Nomada ovata</i>	na	na	na	na	na	na	na	1.459	2
	<i>Osmia bucephala</i>	<i>Osmia cornifrons</i>	0.76	7.1 0	13.5	na	na	na	2.866 67	1
	<i>Osmia cornifrons</i>	<i>Osmia cornifrons</i>	0.76	7.1 0	11.5	na	na	na	2.322	12
	<i>Osmia lignaria</i>	<i>Osmia cornifrons</i>	0.76	7.1	9.5	na	na	na	2.428	2

				0					8	
	<i>Osmia pumila</i>	<i>Osmia cornifrons</i>	0.76	7.1 0	7.5	na	na	na	1.622 88	1
	<i>Xylocopa virginica</i>	male <i>Xylocopa</i> <i>virginica</i>	0.66	2.9 5	17	na	0.82	0.82	6.412 8	114

Figure S1.

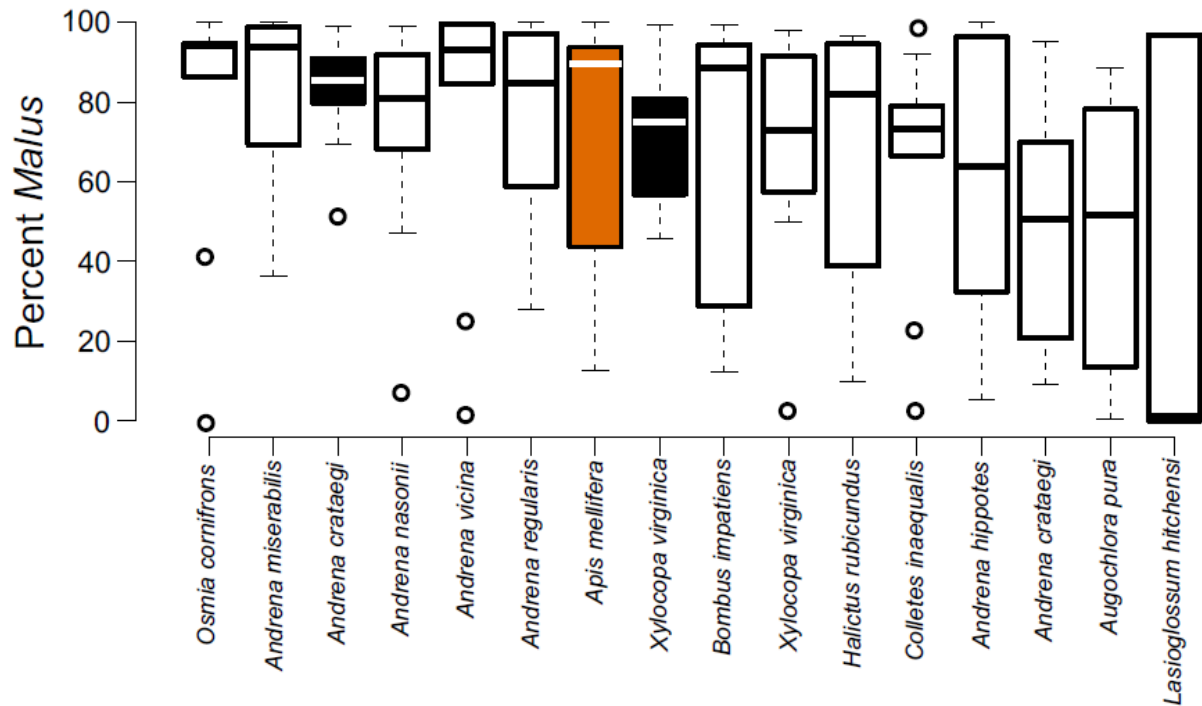


Figure S1. Boxplots of the proportion of *Malus* pollen carried by the 10 individual female of 13 species of bees (white bars). Ten individuals of *Apis mellifera* are marked by an orange bar. In addition, we include the proportion of *Malus* carried by ten individual males (black bars) of two species (*Andrena crataegi* and *Xylocopa virginica*) where the males outnumbered the females in our surveys.

Figure S2.

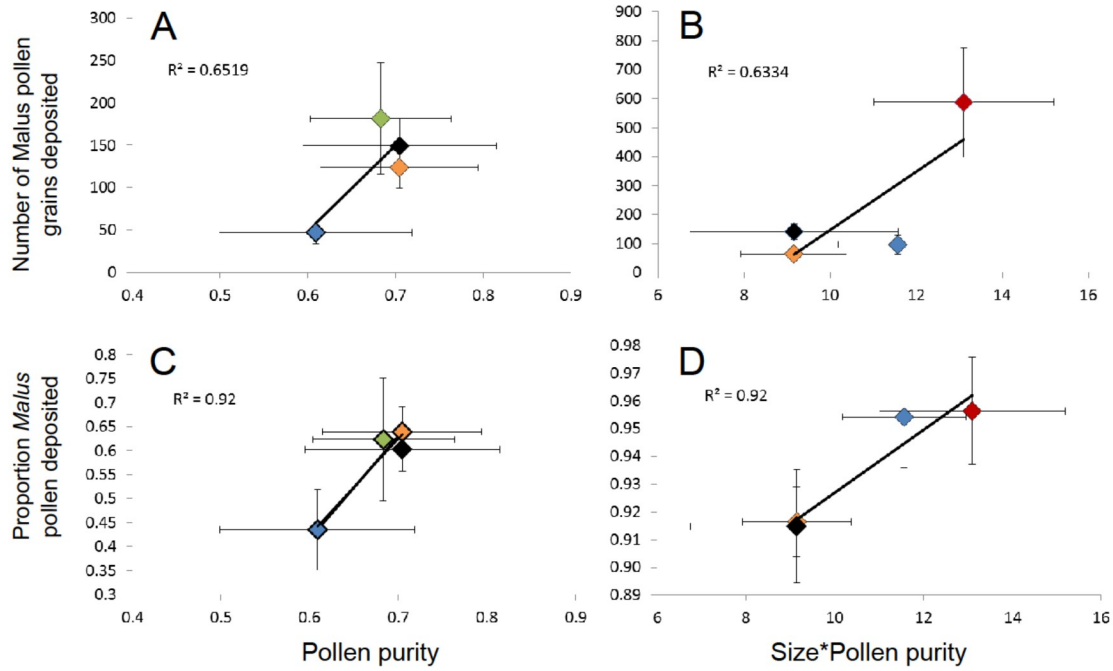


Figure S2. Relationships between size, pollen purity, and pollen deposition by chilled (A, C) and interviewed (B, D) bees. In this figure, green denotes *Andrena nasonii*, red *Xylocopa virginica*, black bees of the subgenus *Melandrena*, orange *Apis mellifera*, and blue bees of the genus *Bombus*. The error bars represent standard errors for both measures.